SOHAN PATIL

AI/ML Engineer

Binghamton, New York | sohanp655@gmail.com | (607) 232-9305 | LinkedIn

SUMMARY

AI/ML Engineer with 5+ years of experience building scalable, production-grade AI systems across fintech and large-scale consumer platforms. Specialized in LLM-powered applications, Retrieval-Augmented Generation (RAG), and low-latency inference systems serving millions of requests daily. Strong expertise in distributed backend engineering, event-driven architectures, and real-time data pipelines using Kafka, Spark, Kubernetes, and AWS. Proven track record of improving model accuracy, reducing latency, and optimizing compute costs in high-concurrency environments. Collaborative, product-focused engineer who translates complex ML capabilities into secure, reliable, and measurable business impact.

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Stripe **1-2025 – Present | CA, USA**

- Led the production rollout of Stripe's LLM-powered developer agents (code generation, CI failure analysis, automated PR creation), supporting millions of daily inferences, reducing debugging latency by 30%, and improving model response accuracy by 18% in a regulated fintech environment.
- Drove performance optimization initiatives (async batching, caching, streaming inference) that delivered sub-second feedback loops and reduced compute costs by 25%, enabling AI tools to scale reliably across thousands of internal engineers.
- Helped operationalize AI into core engineering workflows, automating 40% of repetitive operational tasks and significantly reducing CI investigation time, transforming the platform from experimentation to a trusted, production-grade system.
- Designed and built LLM inference microservices using Python (FastAPI, asyncio, gRPC) deployed on Kubernetes, implementing Ray Serve–style scalable serving patterns, model versioning, and zero-downtime rollbacks.
- Engineered a modular RAG-based architecture integrating GPT-style models with semantic retrieval (FAISS/Pinecone), embedding pipelines, and context enrichment to generate repository-aware, high-precision code suggestions.
- Developed event-driven AI orchestration pipelines using Kafka, chaining multi-step agent workflows (Generate → Validate → Correct → Deploy) with retry logic, idempotency controls, timeout handling, and deterministic guardrails.
- Built robust data and feature engineering pipelines leveraging Spark and Airflow to process logs, stack traces, and CI artifacts; stored structured data in SQL/NoSQL systems and embeddings in vector databases to power context-aware inference.
- Implemented end-to-end MLOps practices using MLflow and CI/CD for ML, enabling controlled experimentation, A/B testing, canary releases, model monitoring, and rapid rollback across cloud-native environments (AWS).
- Strengthened AI reliability and compliance by adding prompt injection defenses, output validation layers, RBAC enforcement, and audit logging to prevent unsafe actions and sensitive financial data leakage.
- Owned AI observability and production support, instrumenting monitoring for inference latency, drift detection, error rates, and throughput; collaborated cross-functionally with DevOps and security teams to maintain high availability under peak load.

Python Full Stack (AI/ML), Meta **8-2019 to 7-2023 | India**

- Built and optimized multilingual NLP pipelines using Python and PyTorch to support translation models covering 200+ languages, enabling scalable deployment of Indian language AI services with improved inference efficiency and model accuracy.
- Integrated open-source LLMs including LLaMA, NLLB-200, and MMS into backend services, developing RESTful APIs with Django and FastAPI to power multilingual text and speech processing applications at scale.
- Developed end-to-end ML workflows for training, validation, and deployment of translation and speech models, leveraging PyTorch and distributed data pipelines to improve processing throughput and reduce model latency.
- Contributed to large multilingual dataset engineering efforts, preprocessing and curating data for low-resource Indian languages, increasing training data quality and improving translation performance across regional language benchmarks.
- Designed backend microservices supporting AI-driven features for social platforms, implementing scalable Python services that handled high concurrent requests while maintaining reliability and optimized response times.
- Optimized model serving performance through batching, caching, and memory tuning strategies, reducing inference latency and improving resource utilization across internal ML infrastructure.
- Collaborated with cross-functional research and product teams to integrate multilingual AI capabilities into user-facing features, aligning technical delivery with product objectives and regional expansion goals.
- Implemented CI/CD pipelines and containerized ML services using Docker, ensuring consistent deployment workflows and improving release stability for AI-driven backend systems.

EDUCATION

State University of New York at Binghamton, New York, United States
Master of Science in Computer Science
Shri Guru Gobind Singhji Institute of Engineering & Technology, Nanded
Bachelor of Information Technology (B. Tech)

SKILLS

Programming & Backend Engineering
Python (AsyncIO, FastAPI, Django, gRPC), SQL, Scala (Spark), Java (microservices), RESTful APIs, Event-Driven Architecture, Distributed Systems Design
AI/ML & Generative AI Engineering
Large Language Models (GPT-style, LLaMA, NLLB-200, MMS), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings & Semantic Search, LLM Orchestration & Agent Workflows, Hallucination Mitigation, Model Evaluation Frameworks, Fine-Tuning, Multilingual NLP, Speech AI
Model Serving & Inference Systems
FastAPI-based Inference APIs, Ray Serve Patterns, TorchServe, NVIDIA Triton, Async Batching & Caching, Streaming Inference, Low-Latency Optimization, Model Versioning & Rollbacks
Data Engineering & Streaming
Apache Kafka, Apache Spark, Spark Structured Streaming, Apache Airflow, ETL Pipelines, Feature Engineering, Real-Time Event Processing, Data Versioning, Parquet
Vector Search & Retrieval Infrastructure
FAISS, Pinecone, Vector Indexing, Hybrid Search (BM25 + Dense Retrieval), Context Enrichment Pipelines, Redis Caching
Cloud & Infrastructure
AWS (S3, EC2, DynamoDB, Redshift, SageMaker, Lambda), Kubernetes, Docker, Terraform, GPU Compute Infrastructure, Autoscaling Systems
MLOps & Production AI Systems
MLflow, CI/CD for ML, Canary & Blue-Green Deployments, A/B Testing, Experiment Tracking, Drift Detection, Monitoring & Observability (Prometheus, Grafana), Audit Logging & Compliance Controls
Security & Reliability Engineering
RBAC, Prompt Injection Mitigation, Secure Model Pipelines, Fault-Tolerant Systems, Exactly-Once Semantics, Backpressure Handling, High Availability Architecture
Collaboration & Ownership
Cross-Functional Delivery, Production On-Call Ownership, Technical Documentation, Stakeholder Communication, Product-Aligned AI Development